

sxchan: An R Package for Channel Cross-Section Analysis

2025-08-19

Summary

sxchan is an R package that helps researchers analyze river channels by creating and manipulating cross-sectional data. River channels can be described in two ways: as horizontal lines across the channel (planimetric cross-sections) and as elevation profiles along those lines (profile cross-sections). The package allows users to generate these cross-sections from various data sources including river bank outlines, digital elevation models, or simple width specifications. Once created, users can erode the banks to create newly eroded channels. Channel properties can be calculated, like width and depth, centerlines generated and banklines traced. The package integrates with existing R spatial analysis tools, making it easy to incorporate into larger research workflows for studying river processes, erosion, and channel evolution in a programmatic way.

Statement of Need

River channel analysis requires handling both planimetric cross-sections (horizontal line segments extending bank-to-bank) and profile cross-sections (elevation data along those segments). Current approaches often require manual GIS workflows or separate tools that don't integrate well with the R ecosystem or automation. Existing R packages for spatial analysis (e.g., **sf** (Pebesma 2018), **terra** (Hijmans 2023)) provide excellent tools for individual operations but lack specialized functionality for channel cross-section analysis.

The advent of remote sensing technologies, particularly airborne LiDAR and photogrammetry, has provided the ability to capture high-resolution topographic data over large spatial extents (Milan et al. 2011; Westaway, Lane, and Hicks 2003). These technologies enable the generation of digital elevation models (DEMs) that can be used to extract detailed channel and floodplain geometry information. However, the challenge lies in efficiently processing these data to extract meaningful cross-sectional information and integrate it with other spatial datasets.

Current approaches to channel analysis rely on manual GIS workflows using software such as ArcGIS or QGIS, where users must manually digitize cross-sections, extract elevation profiles, and perform geometric calculations through point-and-click interfaces. These workflows are time-consuming, error-prone, and difficult to reproduce. While R packages like **sf** and **terra** provide excellent tools for spatial data manipulation and raster processing, they lack specialized functionality for channel cross-section analysis. Researchers must cobble together multiple packages and custom scripts to achieve what should be a unified workflow: generating cross-sections from banklines, sampling DEMs for elevation data, and analyzing channel geometry properties.

This gap is particularly evident in the context of fluvial geomorphology research, where researchers need to analyze channel geometry changes over time, assess erosion and deposition patterns, and model channel evolution processes (Church 2006; Eaton, Church, and Millar 2010). The **sxchan** package addresses this gap by providing a unified framework for channel cross-section analysis that integrates seamlessly with the existing R spatial analysis ecosystem.

Key Features

The package provides a unified data structure for channel objects that integrates planimetric and profile cross-sections. Plan-view cross-sections extend from bank to bank, while profile cross-sections are elevation

profiles along those lines. To enable informed erosion modeling into the floodplain, profile cross-sections can extend beyond the banks into the surrounding topography.

Data Structures

The package deals with two main object types:

- **Channels:** The primary objects, comprised of cross-sections
- **Profile cross-sections:** Specialized geometry objects requiring custom handling

Plan-view cross-sections are represented as simple line segments (using `sf` geometries), avoiding unnecessary complexity and keeping the package lightweight.

Channel Objects

Channels are implemented as data frames (or tibbles (Müller and Wickham 2020) when available), with plan and profile cross-sections stored as columns. Additional columns can represent channel properties at each cross-section, following the design philosophy of the `sf` package where spatial objects are just enhanced data frames.

Profile Cross-Sections

Profile cross-sections are specified as coordinate matrices with distance along the profile in the first column and elevation in the second. They encode bank locations and thalweg points (potentially multiple thalwegs for complex channels).

Design Philosophy

The package prioritizes human readability and consistency. Functions are prefixed with `xt_` (for “cross-section type”) to avoid conflicts and improve code clarity, following the pattern established by the `sf` package. The `units` package ensures consistent dimensional analysis throughout.

Channel Widening

A key functionality is channel widening, which simulates bank erosion processes. For plan-view cross-sections, widening simply extends the line segments. For profile cross-sections, the approach preserves the existing profile while widening the channel from the thalweg outward. This means the channel bottom (thalweg) becomes wider, with the river section between the bank and thalweg shifting laterally, and corresponding beyond-bank topography being removed.

Erosion Allocation

Channel widening operations require specifying how erosion is distributed between left and right banks. The package implements a flexible splitter system inspired by the `scales` package’s transform objects (Wickham and Seidel 2016), providing both simple options (e.g., “left”, “right”, “both”) and extensible custom allocation schemes. Built-in splitters handle common cases, while the `new_splitter()` function enables users to create sophisticated erosion patterns based on bank properties, channel geometry, or environmental factors. This design ensures the API remains simple for basic use cases while providing unlimited flexibility for advanced applications.

Example Usage

```
library(sxchan)

# Load data
banks <- sf::st_read("river_banks.gpkg")
```

```

dem <- terra::rast("elevation.tif")

# Generate channel planimetric cross-sections from demo bankline data
channel <- xt_channel_from_banks(demo_bankline, n = 100)
plot(demo_bankline, col = "lightblue")
plot(channel, view = "plan", add = TRUE)

# Sample DEM for profile cross-sections
plot(demo_dem)
channel <- xt_sample_dem(dem, extent_multiplier = 2, sample_n = 100) |>
  xt_dredge_rectangle(depth = units::set_units(5, "m"))
plot(channel, view = "3d")

# Widen the channel
channel2 <- xt_widen_channel(channel, width = units::set_units(10, "m"))
plot(channel2, view = "3d")

# Analyze results
heights <- xt_height(channel, .f = min)
widths <- xt_width(channel)

```

Acknowledgments

The authors thank BGC Engineering Inc. for funding this work. We also thank Brett Eaton for guiding the package utility, and Heba Abdelmoaty for contributing to algorithm design and the developers of the `sf`, `terra`, `units`, and `scales` packages for providing the foundational tools and design inspiration that make `sxchan` possible.

References

- Church, Michael. 2006. "Bed Material Transport and the Morphology of Alluvial River Channels." *Annual Review of Earth and Planetary Sciences* 34: 325–54.
- Eaton, Brett C, Michael Church, and Robert G Millar. 2010. "Predicting Downstream Hydraulic Geometry: A Test of Rational Regime Theory." *Journal of Geophysical Research: Earth Surface* 115 (F4).
- Hijmans, Robert J. 2023. "Terra: Spatial Data Analysis." *R Package Version 1* (7): 71.
- Milan, David J, George L Heritage, Andrew RG Large, and Neil S Entwistle. 2011. "Terrestrial Laser Scan-Derived Topographic and Roughness Data for Hydraulic Modelling of Gravel-Bed Rivers." *Earth Surface Processes and Landforms* 36 (10): 1334–48.
- Müller, Kirill, and Hadley Wickham. 2020. "Tibble: Simple Data Frames." *R Package Version 3* (1): 1.
- Pebesma, Edzer. 2018. "Simple Features for r: Standardized Support for Spatial Vector Data." *The R Journal* 10 (1): 439–46.
- Westaway, Richard M, Stuart N Lane, and D Murray Hicks. 2003. "Remote Sensing of Clear-Water, Shallow, Gravel-Bed Rivers Using Digital Photogrammetry." *Photogrammetric Engineering & Remote Sensing* 69 (11): 1273–84.
- Wickham, Hadley, and Dana Seidel. 2016. "Scales: Scale Functions for Visualization." *R Package Version 1* (2): 1.